Separable Differential Equations

MATH 146 – Calculus II

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Differential Equations

Differential Equations

A **differential equation** is an equation that relates an unknown function $y(x)$ and one or more of its derivatives.

A differential equation with an initial condition $y(x_0) = y_0$ is an **initial value problem**.

Example: Differential Equations

Example: If $F'(x) = 6x^3$, find $F(x)$.

Example: If $F'(x) = 6x^3$, find $F(x)$ such that $F(1) = 3$.

Example: A car is traveling at the rate of $88\,{}^{ft}\!/_{sec}$ when the brakes are applied. The car begins decelerating at a constant rate of $15\,{}^{ft}\!/_{sec^2}$.

1. How many seconds before the car stops?
2. How far does the car travel during that time?

Example: A rock is thrown upward from the surface of the earth with an initial velocity of $3\,{}^{m}\!/\!{}_{s}$ from a height of 2 meters. When does it reach its highest point?

Recall: Objects on the surface of the earth accelerate toward the center of the earth at a rate of $9.8\,{}^{m}\!/\!{}_{s^2}$.

Separable Differential Equations

Separable Differential Equations

A first order differential equation is said to be **separable** if it can be written in the form $y' = f(x)g(y)$.

Such an equation can (technically) be solved using substitution of integrals.

Example: Separable Differential Equations

Example: Consider the initial value problem $y' = x^2 y$, $y(0) = 3$.

Example: Separable Differential Equations

Example: Consider the IVP $\dfrac{dx}{dy} = -\dfrac{x}{y}$, $y(4) = -3$.

Example: Separable Differential Equations

Example: Consider the IVP $y' = ye^{-x^2}$, $y(0) = 1$.

Example: Separable Differential Equations

Example: Consider the differential equation $y' = xy + x + y + 1$.